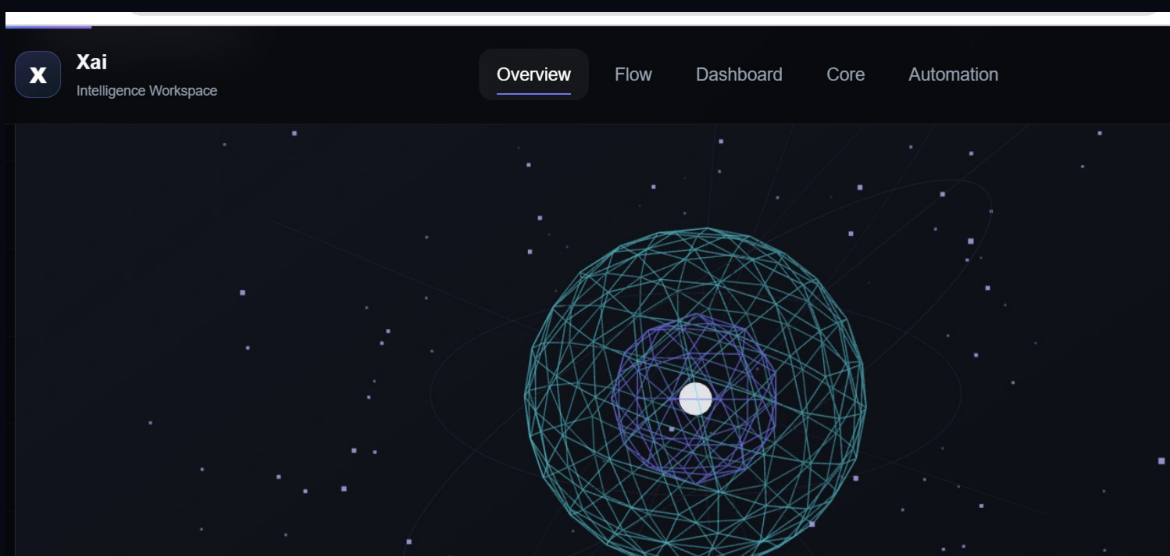
XAI / INTELLIGENCE WORKSPACE

From complex data to confident decisions.

Product documentation and experience rationale

Raw data → Structured intelligence → Actionable insight → AI automation



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Frontend Product Experience · 2026

Executive Summary

A product-quality frontend prototype designed to communicate how Xai turns fragmented information into coordinated action.

> **The central idea**
> Xai is not presented as a marketing landing page. It is designed as a calm, technically confident intelligence workspace for decision-makers.

Product idea

Xai connects scattered business signals, analyzes their relationships, identifies meaningful risk and opportunity patterns, and turns those patterns into reviewable actions. The experience uses motion, geometry, dashboard states, and automation feedback to make that transformation visible and understandable.

Core scenario

18%	14	$420K	94%
Renewal risk increase	High-value accounts	Revenue exposure	Model confidence

Design principles

Clarity before spectacle - every visual element supports product understanding.

Motion with intent - animation communicates transitions in data, confidence, and workflow state.

Enterprise restraint - dark neutral surfaces, precise spacing, calm typography, and disciplined color.

Inspectable intelligence - evidence, impact, and confidence are available before action is approved.

Human control - automation remains visible, reviewable, pausable, and accountable.

Experience Architecture

A single-page journey that moves from abstract signals to a monitored business response.

NARRATIVE SEQUENCE

01	02	03	04
Raw signals	**Structured intelligence**	**Actionable insight**	**AI automation**
Scattered particles and orbiting geometry represent disconnected business data.	The data organizes into a coherent model and a three-stage product flow.	The dashboard and Intelligence Core explain risk, confidence, and business impact.	The user reviews, approves, monitors, pauses, and resumes a coordinated workflow.

Section map

Section	Purpose	Primary interaction
Overview	Introduces the product promise and the signature 3D intelligence field.	Scroll and cursor response
Flow	Explains Ingest Data, Analyze with AI, and Generate Insight.	Progressive reveal and card interactions
Dashboard	Presents metrics, forecast, risks, opportunities, and prioritized intelligence.	Tabs, ranges, hover and state changes
Core	Lets users inspect signal clusters and supporting evidence.	3D cluster selection and synchronized detail
Automation	Turns a recommendation into a controlled workflow.	Review, approve, execute, pause, resume and audit

Interaction and Motion System

Motion is treated as product communication, not decoration.

Signature 3D interaction

The hero uses particles, wireframe geometry, orbiting structures, depth, pointer response, and subtle floating motion to represent raw data becoming structured intelligence. The visual is intentionally abstract: it communicates system behavior without relying on stock illustrations, Lottie files, or external 3D models.

> **WOW moment**
>
> The Intelligence Core provides a deliberate, high-value interaction: users select a cluster and immediately see the corresponding evidence, confidence, estimated impact, and recommended action.

Motion specification

Interaction	Trigger	Animation	Duration	Easing
Hero particles	Scroll / cursor	Drift, organize, and respond in depth	Continuous / 800ms	0.16, 1, 0.3, 1
Core float	Continuous	Vertical float, -16px to 0px	6s loop	Ease in-out
Orbit rings	Continuous	Counter-rotating rings	6s loop	Linear
Inner core	Continuous	Scale pulse, 1 to 1.12 to 1	6s loop	Ease in-out
Pipeline cards	Enter viewport	Fade and rise with stagger	600ms	0.16, 1, 0.3, 1
Dashboard tabs	Tab change	Crossfade and content state update	300ms	0.2, 0, 0, 1
Cluster selection	Click / keyboard	Emphasize active cluster; dim others	500ms	0.16, 1, 0.3, 1
Workflow progress	State update	Progress bar and step transition	400ms	Linear

Interface and Design System

A compact visual language built for calm, high-confidence decision-making.

Visual direction: Controlled Intelligence

The design combines dark enterprise surfaces with restrained violet and cyan accents. Large areas remain neutral so risk, opportunity, confidence, and progress states can be understood immediately. The overall tone is technical and premium without becoming visually noisy.

Core palette

#070A12	#111625	#8B7CFF	#5FE3E8	#63D69B	#F1B55B
Foundation Primary background	**Surface** Panels and navigation	**Violet** Intelligence and active states	**Cyan** Structure and depth	**Success** Live and completed states	**Warning** Exposure and attention

Component strategy

- Reusable navigation, section heading, metric card, workflow step, evidence panel, and table patterns.
- Feature-owned modules keep dashboard, core, automation, flow, and effects logic separate.
- Consistent radius, spacing, typography, focus, and semantic status treatments.
- Desktop-first composition with responsive stacking and simplified mobile hierarchy.
- Static demonstration data is used intentionally; no backend is required for the challenge.

Engineering Architecture

A modular Next.js implementation with purposeful animation boundaries.

Technology stack

Technology	Responsibility
Next.js + React	Application structure and rendering
TypeScript	Type safety and maintainable state
Framer Motion	UI entrances, state transitions, and choreography
GSAP	Advanced scroll and timeline capability
Three.js / React Three Fiber	Meaningful 3D scenes in the hero and Intelligence Core
Lucide React	Consistent interface iconography
Playwright	Experience-level test starter

Component architecture

The codebase is organized by feature ownership rather than placing the complete experience in one oversized page component.

- components/intelligence-flow - pipeline cards and data-to-insight visuals
- components/dashboard - dashboard data, metrics, chart, evidence, and table interactions
- components/core - Intelligence Core section and cluster-selection state
- components/intelligence-core - WebGL / Three.js core scene
- components/automation - review, approval, execution, pause, reset, and audit trail
- components/effects - shared scroll progress and visual loading states
- components/layout - navigation, hero composition, and page assembly

> **Engineering discipline**
>
> Dynamic 3D imports, limited device pixel ratio, lightweight generated geometry, deterministic particles, and minimal post-processing reduce unnecessary rendering cost.

Accessibility, Responsiveness and Performance

The visual experience remains usable when motion, input method, or viewport changes.

Accessibility decisions

- Semantic HTML structure and a clear heading hierarchy.
- Keyboard-accessible buttons, tabs, cards, and cluster controls.
- Visible focus states and descriptive control labels.
- Sufficient contrast across dark surfaces and semantic status colors.
- Reduced-motion support that removes unnecessary continuous movement.
- Text-based alternatives for important information shown in 3D scenes.
- Tables and metrics remain readable without relying on color alone.

Responsive strategy

Desktop screens preserve the full product narrative and visual depth. On smaller screens, the layout prioritizes content order: headline, key metrics, primary insight, selected evidence, and workflow status. Complex panels stack vertically, and large 3D scenes are simplified rather than merely scaled down.

Performance strategy

Area	Decision
3D loading	Scenes are loaded dynamically instead of blocking the first render.
Rendering	Device pixel ratio is limited and geometry remains lightweight.
Animation	Transforms and opacity are favored for smooth GPU-friendly transitions.
Assets	No stock images, Lottie files, or external 3D models are required.
Data	Static mock data avoids unnecessary requests and keeps the prototype reliable.
Fallbacks	Loading states and text alternatives protect usability when WebGL is delayed or unavailable.

Validation and Submission

A concise handoff for reviewers and engineers.

Local setup

```
npm install
npm run dev

# validation
npm run typecheck
npm run lint
npm run build
```

Submission links

GitHub repository: https://github.com/ssimuni/xai-intelligence-workspace

Figma design: https://www.figma.com/design/yqJ8L3HoZRu4GTbmVijW8R/Untitled

Live deployment: ADD YOUR VERCEL URL

Demo video: ADD YOUR YOUTUBE OR GOOGLE DRIVE URL

Final checklist

Deliverable	Check
Public GitHub repository	Source code, README, and .gitignore are visible.
Public Figma link	Anyone with the link can view the file.
Live deployment	The production URL opens and interactions are smooth.
Product documentation	This PDF/DOCX is uploaded with public viewing access.
Demo video	The video explains the 3D motion and key interaction decisions.
Repository hygiene	node_modules, .next, .vercel, playwright-report, and test-results are not committed.

Submission statement

Xai demonstrates a complete product story: fragmented data becomes structured intelligence, intelligence becomes an explainable insight, and the insight becomes a controlled AI workflow.